

Executive Summary

Predicting Outpatient Consultation Duration — Hangu Outpatient Oncology Clinic

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The Problem

Clinics need to know how long a consultation will take — both to build a workable daily schedule and to estimate what each visit actually costs to deliver. In practice, most clinics fall back on a single generic appointment length for every patient, which leads to some slots running short and others running long, and makes it hard to answer a basic costing question: *how much clinician time does a typical visit really require?*

What We Built

Using two years of real visit records from an outpatient oncology clinic (6,637 consultations), we built two simple prediction tools that use only information already known at the time an appointment is booked — things like whether the patient has visited before, the day and time of the appointment, and whether a cancer diagnosis is on file:

1. **A “long visit” flag** — predicts whether a given appointment is likely to run longer than the clinic’s own typical visit.
2. **A duration estimate** — predicts roughly how many minutes a given appointment will take.

Both tools are available through a simple web form, so a scheduler can enter a few known details about an upcoming visit and get an immediate estimate.

What We Found

- **The “long visit” flag is moderately reliable.** It correctly identifies a long visit as “long” about 6 times out of 10 — clearly better than guessing, but not precise enough to be the only factor in a scheduling decision.
- **The duration estimate is directionally useful, not exact.** On average, it misses the actual visit length by a little over four minutes. That is accurate enough to plan overall clinic capacity, but not precise enough to set an individual appointment slot with confidence.
- **The single biggest driver of visit length, by far, is whether the patient has been seen before — not the day of the week, the time of day, or the diagnosis on record.** Returning patients and first-time patients behave differently enough that this one piece of information matters more than everything else combined.

What This Means in Practice

- **For scheduling:** use the tools to plan overall daily or weekly capacity and staffing levels, rather than to fix the exact length of any single appointment.
- **For costing (Time-Driven Activity-Based Costing):** the clearest, most defensible way to estimate the “time” side of a cost-per-visit calculation is to separate new-patient visits from returning-patient visits — this distinction carries far more real signal than segmenting by day, time slot, or diagnosis category.
- **For future improvement:** the tools currently only see scheduling information, not clinical detail (for example, the specific reason for the visit or its complexity). Capturing even a little more of that detail would likely improve accuracy meaningfully.

Bottom Line

Ordinary scheduling data — visit history, timing, and a basic diagnosis flag — is enough to produce a genuinely useful, if imperfect, estimate of how long a consultation will take and to flag visits at risk of running long. That is enough to support smarter capacity planning and a more evidence-based costing model today, while making clear that these estimates are a planning aid for administrators, not a substitute for clinical judgment on any individual appointment.